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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: MediLink – Smart Blood & Medicine Emergency Network – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT		
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Abstract

MediLink – Smart Blood & Medicine Emergency Network is a web-based healthcare assistance platform designed to provide quick and reliable access to essential medical resources during emergency situations. In critical conditions, patients and their families often face difficulties in finding compatible blood donors or locating urgently required medicines within a short period of time. Traditional methods such as contacting

multiple hospitals, blood banks, pharmacies, or requesting help through social media can be time-consuming and inefficient. MediLink aims to address this problem by providing a centralized digital platform that connects patients, blood donors, volunteers, and pharmacies.

The system allows users to create and manage accounts, register themselves as blood donors or volunteers, search for available blood donors based on blood group and location, and create emergency blood requests. Registered donors can maintain their profiles, update their availability, and provide relevant donation information. During an emergency, users can specify details such as the required blood group, number of units, hospital, location, urgency level, and contact information. The platform maintains the status of each request, allowing it to progress through stages such as open, accepted, in progress, and completed. A volunteer network can further assist in coordinating emergency requests and connecting people who require help with available resources.

MediLink also includes a Medicine Finder module that helps users search for essential medicines and identify pharmacies where those medicines are available. Users can view pharmacy information, medicine availability, location, and contact details. Location-based functionality can be enhanced through Google Maps API integration, allowing users to identify nearby donors and pharmacies and making emergency resource discovery faster and more convenient.

The application is developed using Java, JSP, HTML5, CSS3, and JavaScript for implementing the web interface and application functionality. MySQL is used as the database to securely maintain information related to users, donors, emergency requests, volunteers, medicines, and pharmacies. JDBC (JavaDatabaseConnectivity) provides

communicationn between the Java application and MySQL database, while Tomcat is used to deploy and run the Java web application. REST APIs can be used for communication with external services, including location-based functionality. Git and GitHub are used for source-code management and version control.

The development of MediLink follows the Agile Scrum methodology, with the project divided into multiple sprints covering user authentication, blood donor management, emergency requests, volunteer management, medicine search, pharmacy services, dashboard development, testing, and integration. Jira is used to organize the product backlog, plan sprints, assign development tasks, and track project progress.

The primary objective of MediLink is to reduce the time and effort required to locate critical healthcare resources by bringing multiple emergency-support services into a single platform. By combining blood donor discovery, emergency request management, volunteer coordination, medicine search, pharmacy information, and location-based services, MediLink demonstrates how modern web technologies can be utilized to develop an accessible, organized, and potentially life-saving healthcare support system.

Signature of the Guide HOD